Lab brief: does force speed the change?

Same total mass. Different driving force. F = ma emerges.

12 A

TASK 1 · KEEP THE MASS FIXED

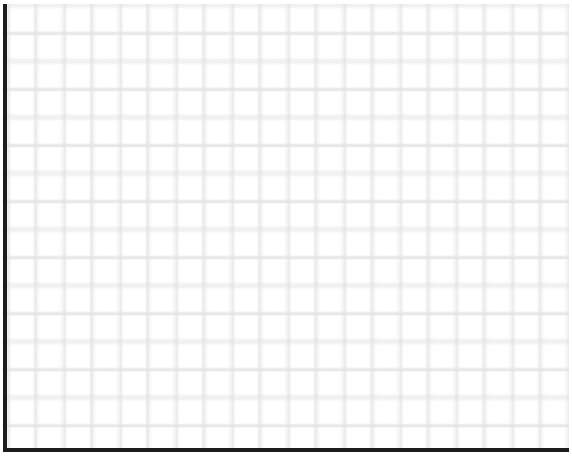
Move masses from the cart to the hanger. Never add new ones. Total moving mass: ____ kg

- ☐ total mass unchanged
- ☐ release from rest
- ☐ track level
- ☐ 2+ trials each

TASK 2 · MY DATA

HANGING MASS (kg)	DRIVING FORCE (N)	TRIAL 1 a	TRIAL 2 a	AVERAGE a

TASK 3 · GRAPH: a vs. FORCE



TASK 4 · READ THE SLOPE

Slope = ____ (m/s²)/N. Mass = 1 ÷ slope = ____ kg. Does it match your total?

ANALYST SIGN-OFF · COMBINE BOTH LABS

a ∝ F and a ∝ 1/m, so a = ____ ÷ ____. My evidence: ____

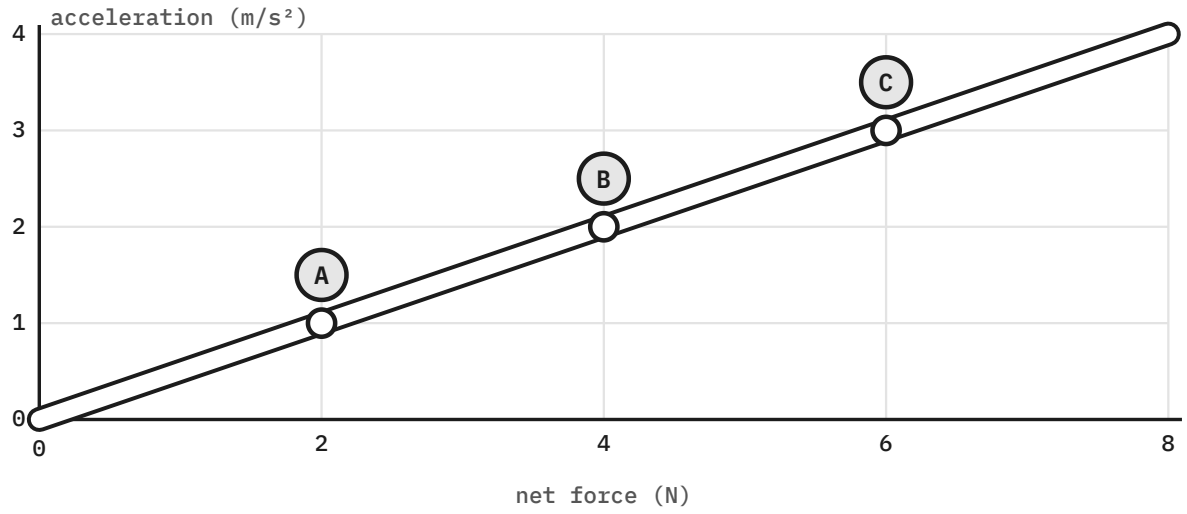
S4 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Read the a vs. F graph

12 F

DIRECTIONS · Ideal 2 kg system (illustrative). Real data may sit lower because of friction.



A · 2 N

a = ____ m/s²

B · 4 N

a = ____ m/s²

C · 6 N

a = ____ m/s²

SLOPE

____ (m/s²)/N → m = ____ kg

PREDICT

Same system, 10 N. a = ____ m/s². What if mass doubled to 4 kg at 10 N? a = ____

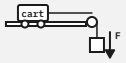
WHY MIGHT REAL DATA MISS THE LINE?

S4 1 2 3

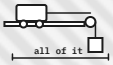
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Say it, then write it: the force lab

Language goal: I can **compare** two results.



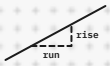
driving force
fuerza impulsora · força motriz
the hanging weight that pulls



total mass
masa total · massa total
cart + hanger + all masses



proportional
proporcional · proporcional
double one, double the other



slope
pendiente · inclinação
rise ÷ run of a graph line

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: force · mass · doubles · proportional · slope

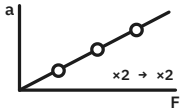
We changed the driving ____ but kept the total ____ the same. When the force doubles, the acceleration ____. So a and F are ____. The ____ of the a vs. F line is 1 ÷ m.

2 · SAY IT TO A PARTNER, THEN WRITE IT



When the force was ____ N, the acceleration was ____ m/s².
When the force doubled, a ____, so they are ____ .

3 · SOLVE WITH THE STEPS



Same 2 kg system: 2 N gives 1 m/s². Predict a at 6 N.

STEP 1 · COMPARE
6 N ÷ 2 N = × ____

STEP 2 · SCALE a
1 × ____ = ____ m/s²

STEP 3 · SAY IT
"Three times the force gives ____ the acceleration."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ Acceleration is proportional to force because ____ .
☐ We moved masses to the hanger so that ____ .
☐ My data point at ____ N was off because ____ .

I said it ☐ · I wrote it ☐

S4 1 2 3

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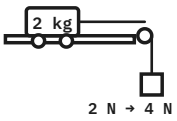
UNIT 1 · DAY 12 · S4 · SEI · PARTNER TALK

Talk it through: the force lab

Language goal: I can **compare** two results.

DIRECTIONS · Read your lines aloud. Listen, then write. Switch roles each round.

ROUND 1 · 2 N → 1 m/s², 4 N → ? (SAME MASS)



A ASKS "What happens to a when the force doubles?"

B SAYS "The force goes from ____ to ____, so a goes from 1 to ____ m/s²."

A CHECKS "So you're saying a and F are ____."

ROUND 2 · SWITCH · WHAT DID WE KEEP THE SAME?

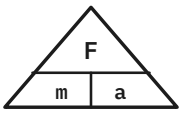


B ASKS "Why did we move masses instead of adding them?"

A SAYS "To keep the ____ the same. If we add mass, ____ changes too."

B ADDS "I want to add that the hanger's ____ is the driving force."

ROUND 3 · BOTH · PUT BOTH LABS TOGETHER



A "More mass gives ____ acceleration."

B "More force gives ____ acceleration. So a = ____ ÷ ____."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

F = ma means…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

S4 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Pitch "PushBot"

Your warehouse robot pushes carts. Investors want to know how fast it can speed them up.

12 Z

SLIDE 1 · TEST DATA (robot pushes with a steady net force)

CART	MASS (kg)	NET FORCE (N)	a = F ÷ m
empty	20	60	
half full	40	60	
full	60	60	
full + boost	60	120	

SLIDE 2 · THE PATTERN

Doubling the mass ____ a. Doubling the force ____ a.

SLIDE 3 · SPEC

A full cart must reach 2 m/s in 2 s. Needed a = ____ . Needed F = ____.

THE PITCH · 30 SECONDS

"PushBot uses F = ma: a heavier cart needs ____ force for the same acceleration. For a full cart we deliver ____ N, so it reaches ____ m/s in ____ s."

SHARK QUESTION

"Why not just push harder on every cart?" Your answer:

S4 1 2 3

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Variable, control, or result?

12 N

DIRECTIONS · Today's lab: change the driving force, keep the total mass. Sort each card.

A hanging mass (driving force)	B total moving mass	C acceleration	D track angle (level)
E velocity–time slope	F release method	G number of masses on the hanger	H start position
I the a vs. F graph	J cart + hanger + all masses	K friction in the setup (try to keep it)	L average a of trials

CHANGED	CONTROLLED	MEASURED
what we vary	kept the same	the result

EXPLAIN ONE TRICKY CARD

Card ____ belongs where it does because _____

S4 1 2 3

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